

# Module 3: Data Organization and Manipulation using Collections

Premanand S

Assistant Professor  
School of Electronics Engineering  
Vellore Institute of Technology  
Chennai campus

*premanand.s@vit.ac.in*

September 7, 2025

## In module 3,

- Data Handling Strategies Using Lists, Tuples, Dictionary and Set
- Data Comprehension
- Data Modification
- Grouping and Categorization
- Iterators
- Data Selection - Ordered, Unordered and Unique Data
- Searching Techniques: Linear Search, Binary Search
- Sorting Techniques: Bubble Sort, Selection Sort- Insertion Sort- Quick Sort- Merge Sort.

# What is a Tuple?

- A tuple is an ordered, immutable collection of elements.
- Defined using parentheses ( ) instead of square brackets [ ].
- Can store mixed data types (int, float, string, list, dict, another tuple, etc.).
- Because tuples are immutable, we cannot modify, add, or delete elements once created.

# What is a Tuple?

- Ordered → Elements have a defined order (indexed).
- Immutable → Cannot change elements once assigned.
- Allow duplicates → Like lists, tuples can contain duplicate values.
- Heterogeneous → Can store multiple data types.
- Indexing & slicing supported.
- Faster than lists because of immutability.

# Tuple creation

## Example (Syntax)

```
fruits = ("apple", "banana", "cherry")  
print(fruits)
```

# Tuple - different DT

## Example (Syntax)

```
mixed = (1, 2.5, "hello", True)
print(mixed)
```

# Tuple - Nested

## Example (Syntax)

```
nested = (1, (2, 3), [4, 5])  
print(nested)
```

# Single element tuple

## Example (Syntax)

```
t1 = (5)
```

```
t2 = (5,)
```

```
print(type(t1))
```

```
print(type(t2))
```



# Tuple without parentheses

## Example (Syntax)

```
t = 1, 2, 3, "python"  
print(t)
```

## Example (Syntax)

```
a = (1, 2)
```

```
b = (3, 4)
```

```
print(a + b)
```

```
nums = (1, 2)
```

```
print(nums * 3)
```

```
t = (10, 20, 30)
```

```
print(20 in t)
```

```
print(50 not in t)
```

## Example (Syntax)

```
print(dir(tuple))
```

```
help(tuple.count)
```

```
help(tuple.index)
```

# Questions to solve

- Create a tuple of 5 student names. Print the **first and last name**.
- Store a tuple of 5 numbers. Find **maximum, minimum, sum**.
- Given `colors = ("red", "green", "blue", "green")`:
  - Count how many times "green" appears.
  - Find the index of "blue".
- What happens if you try to modify an element inside a tuple?

# Questions to solve

- Marks dataset: `marks = (85, 92, 78, 90, 88)`
  - Find the **average marks**.
  - Check if any student scored exactly 100.
- Coordinates: `point = (3, 4)`  
Find the **distance from origin**.
- Tuple of words: `("python", "tuple", "immutable", "fast")`  
Print all words with length  $> 5$ .

# Questions to solve

- Students dataset: `(("Alice",85),("Bob",92),("Charlie",78))`
  - Print student with highest marks.
  - Print names with marks  $> 80$ .
- Sales dataset: `(1200,3000,2500,4000,3500)`
  - Find day with maximum sales.
  - Compute percentage contribution of each day.
- Can a tuple contain a list? Try:  
`t = (1, [2, 3], 4)`  
Modify the list inside and explain why tuples are still immutable.